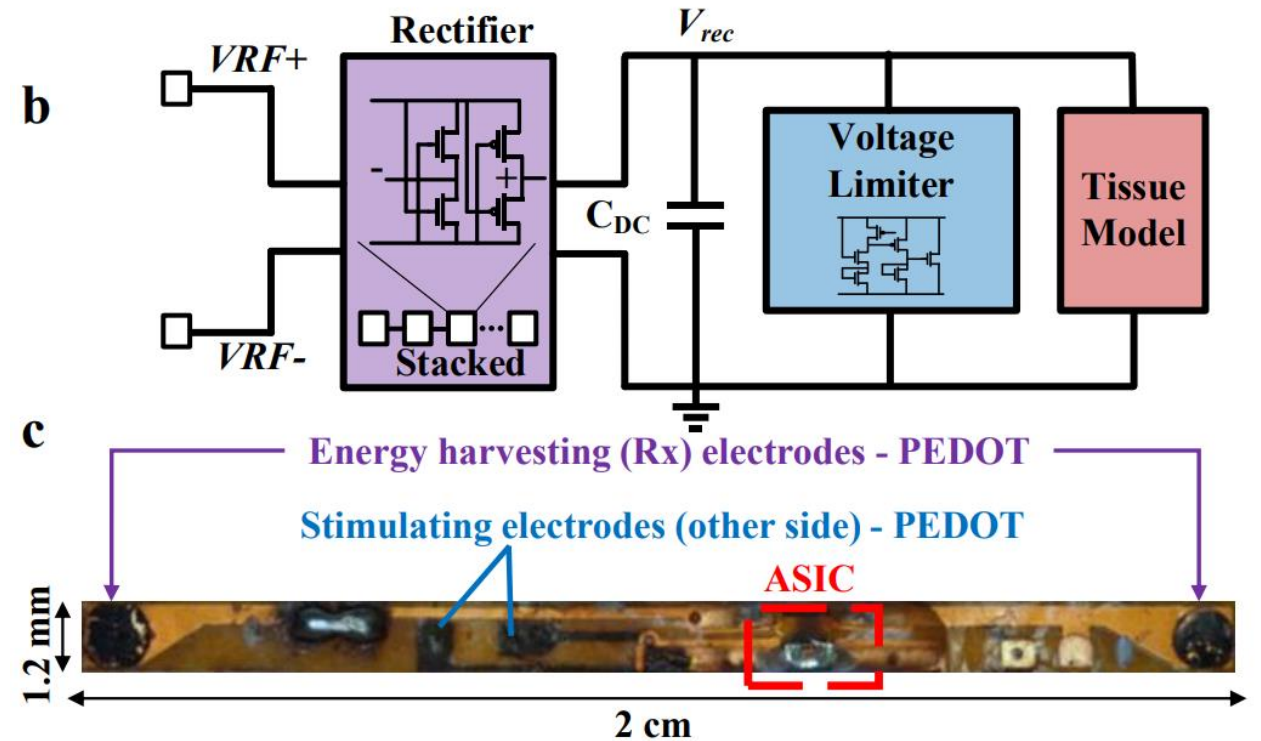
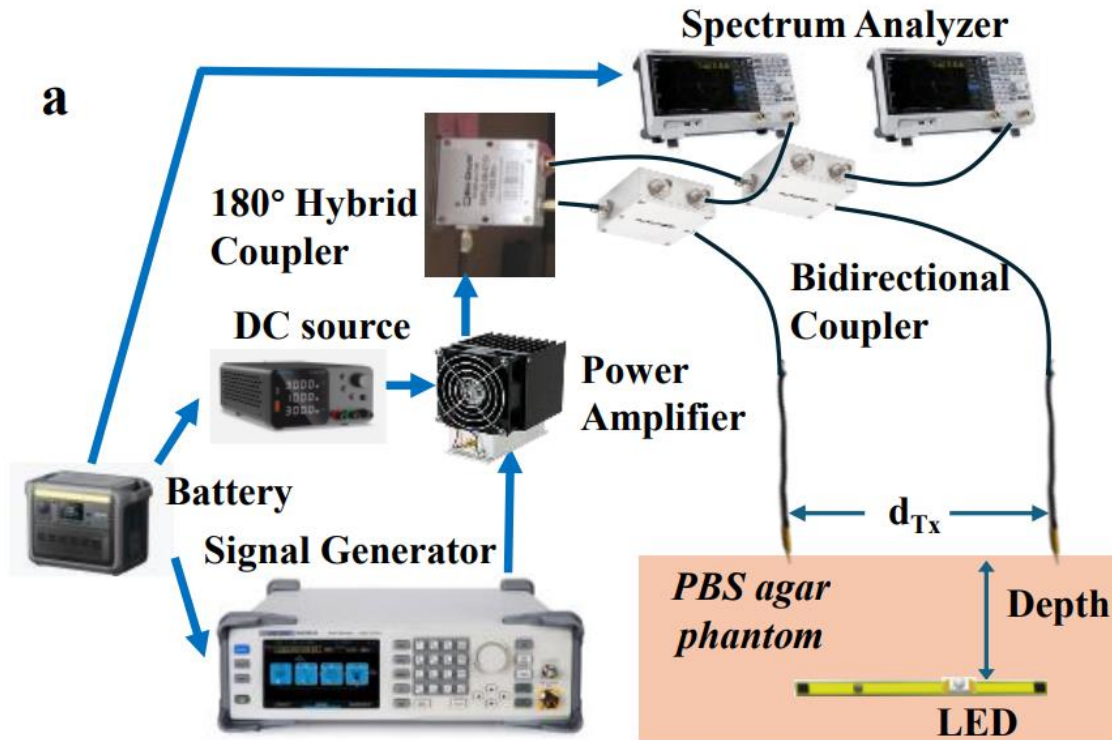


DTCP Prototype Presentation

Jeremiah Dados, Dmytro Stavskyi, Luis Wong

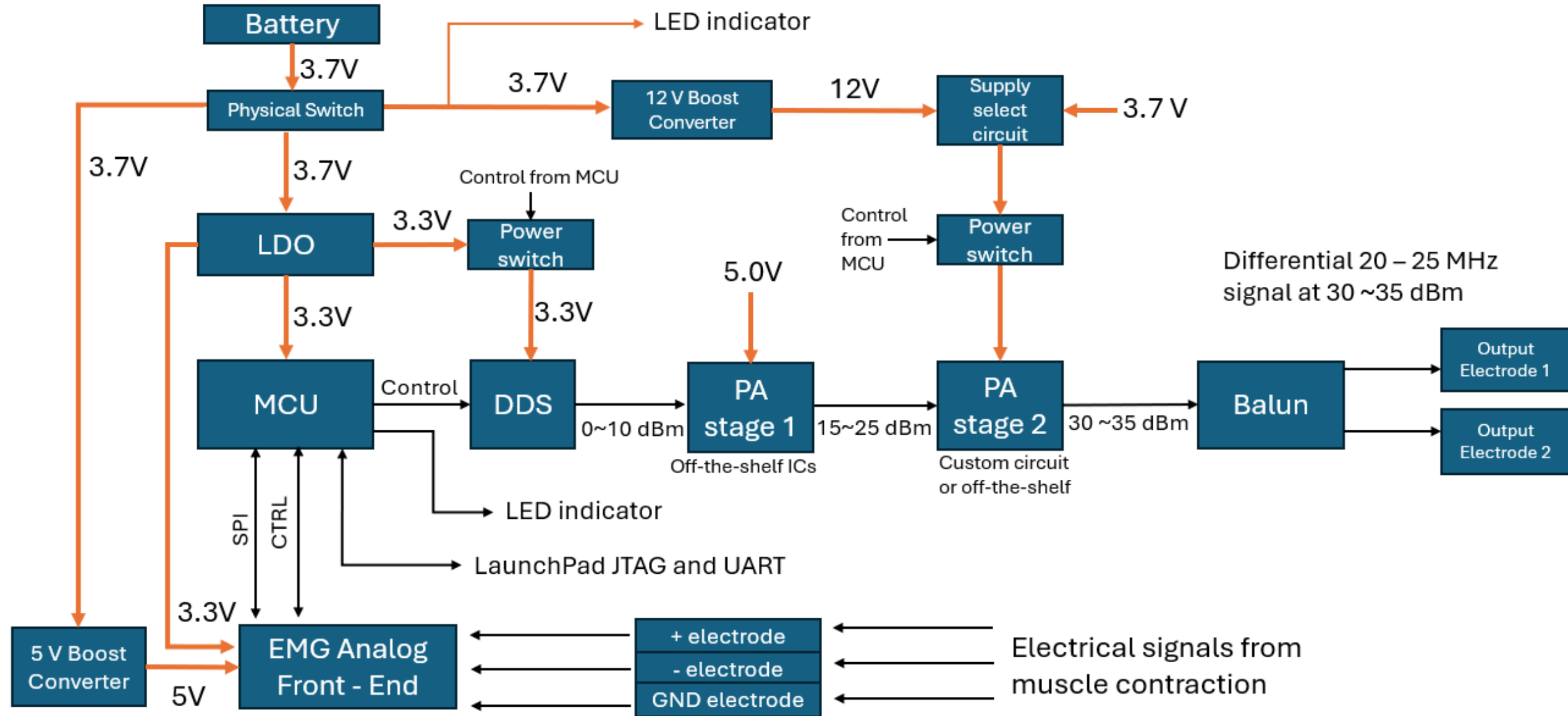
Project objectives

- Build a compact, battery-powered device that can:
 - Output a 500 μ s/pulsed differential 25 MHz signal at 30 dBm or more.
 - Accurately record EMG spikes following DTCP nerve stimulation.

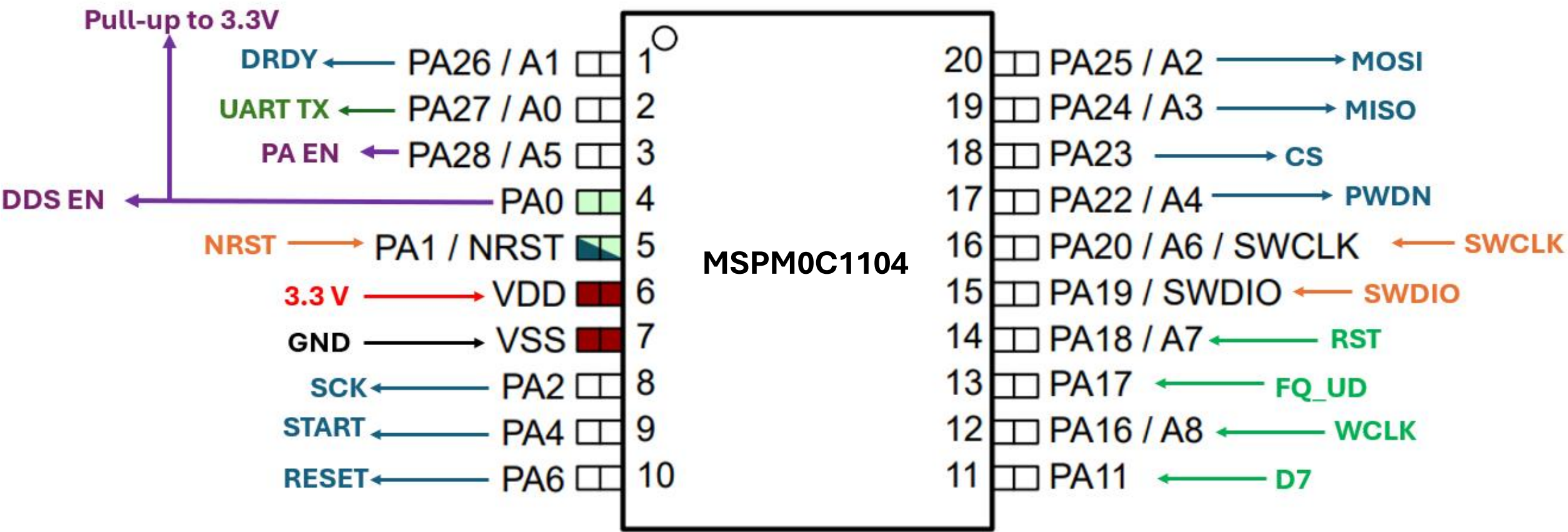


Source: Wu, Han et al. (2025). Differential Tissue Coupled Powering for Battery-Free Injectable Electroceuticals. 10.1101/2025.11.09.687502.

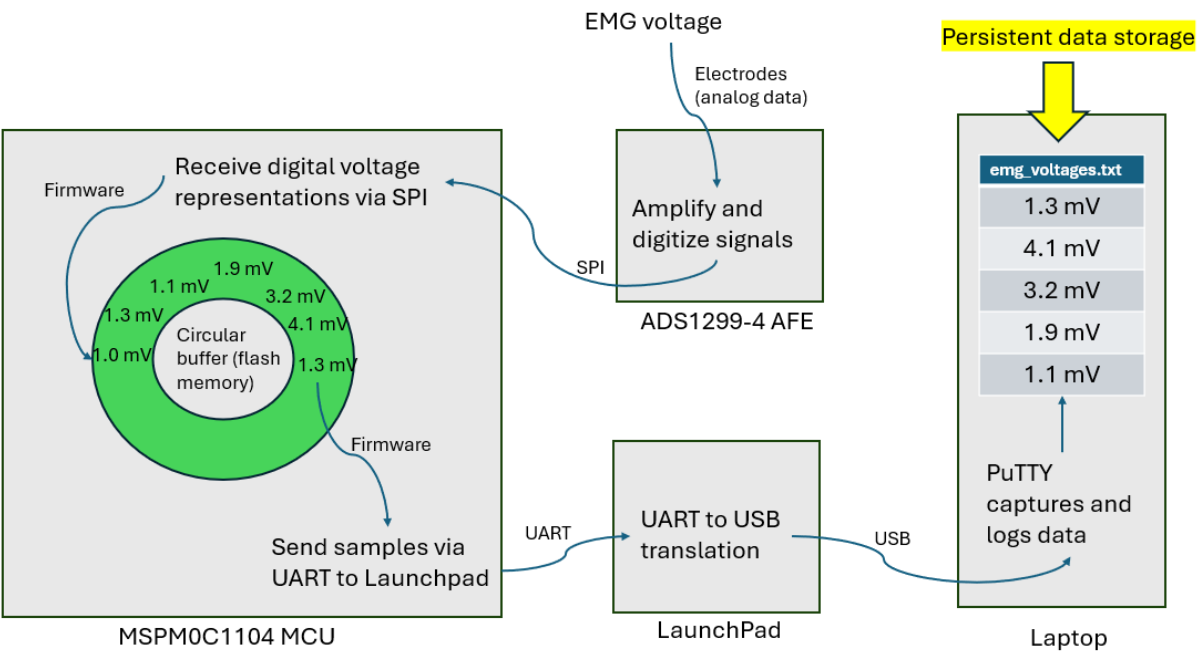
System Architecture



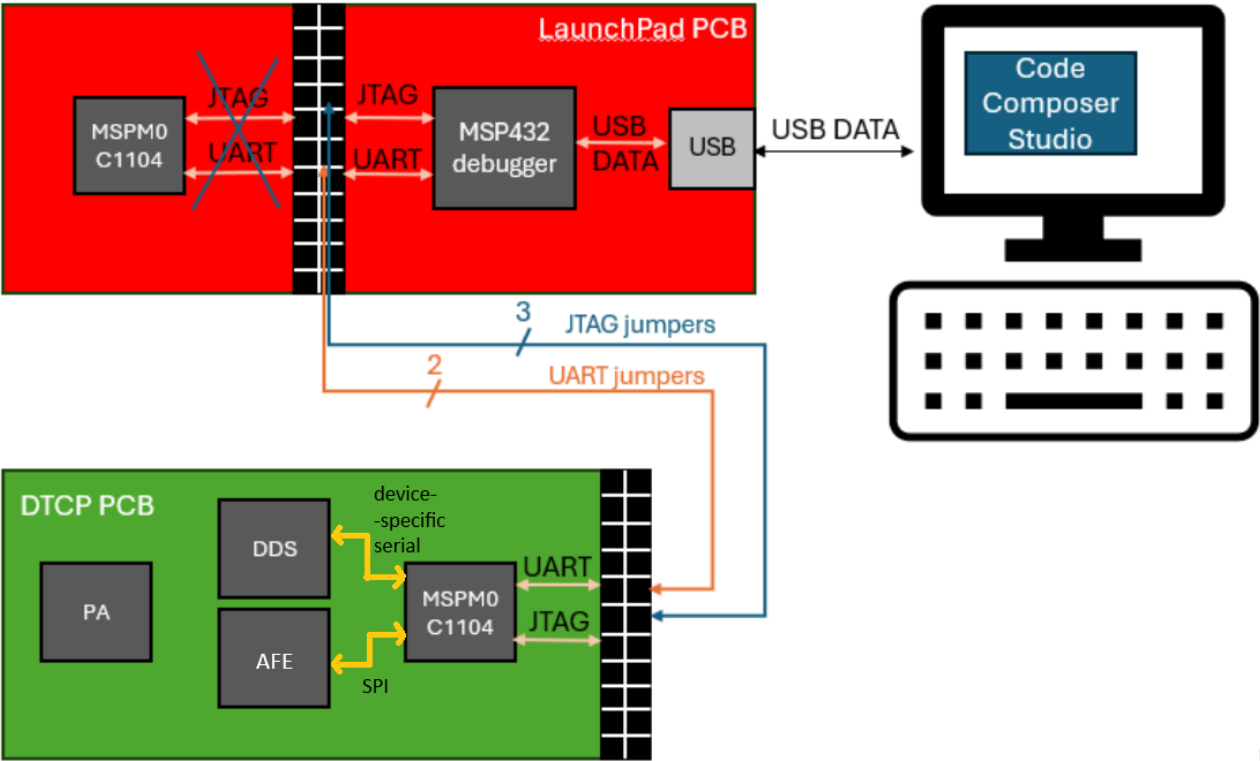
Processing and Control



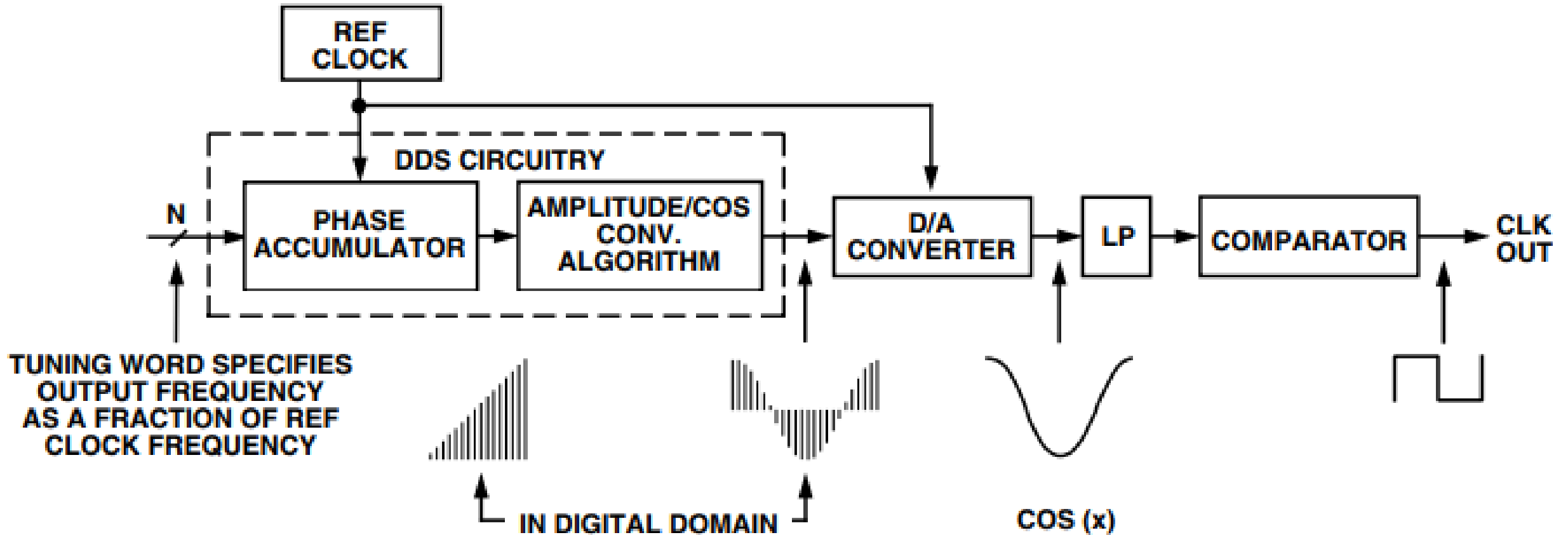
Processing and Control



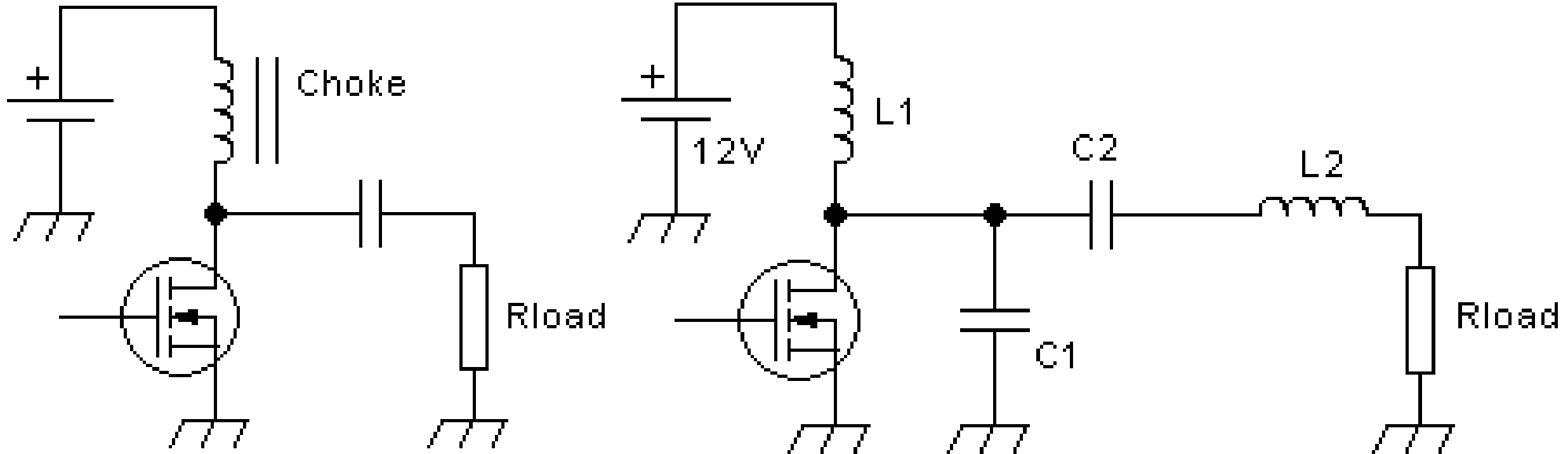
Connection broken by removing the jumpers



RF Signal Generation

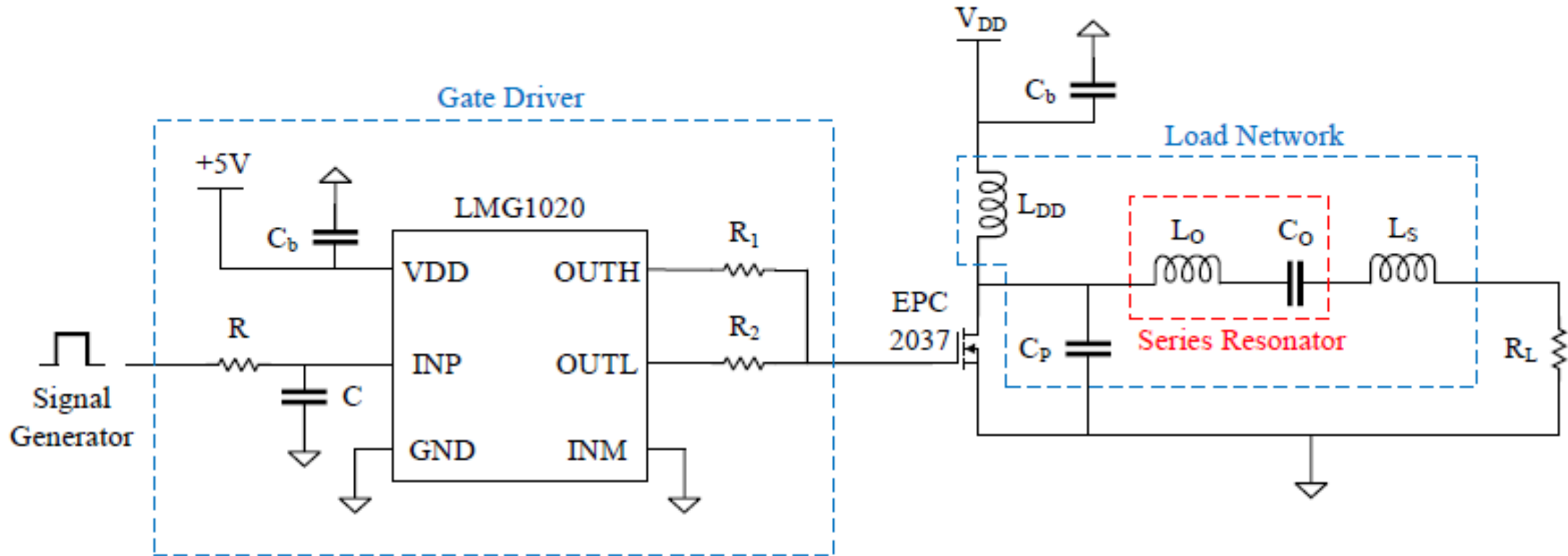


RF Power Amplification



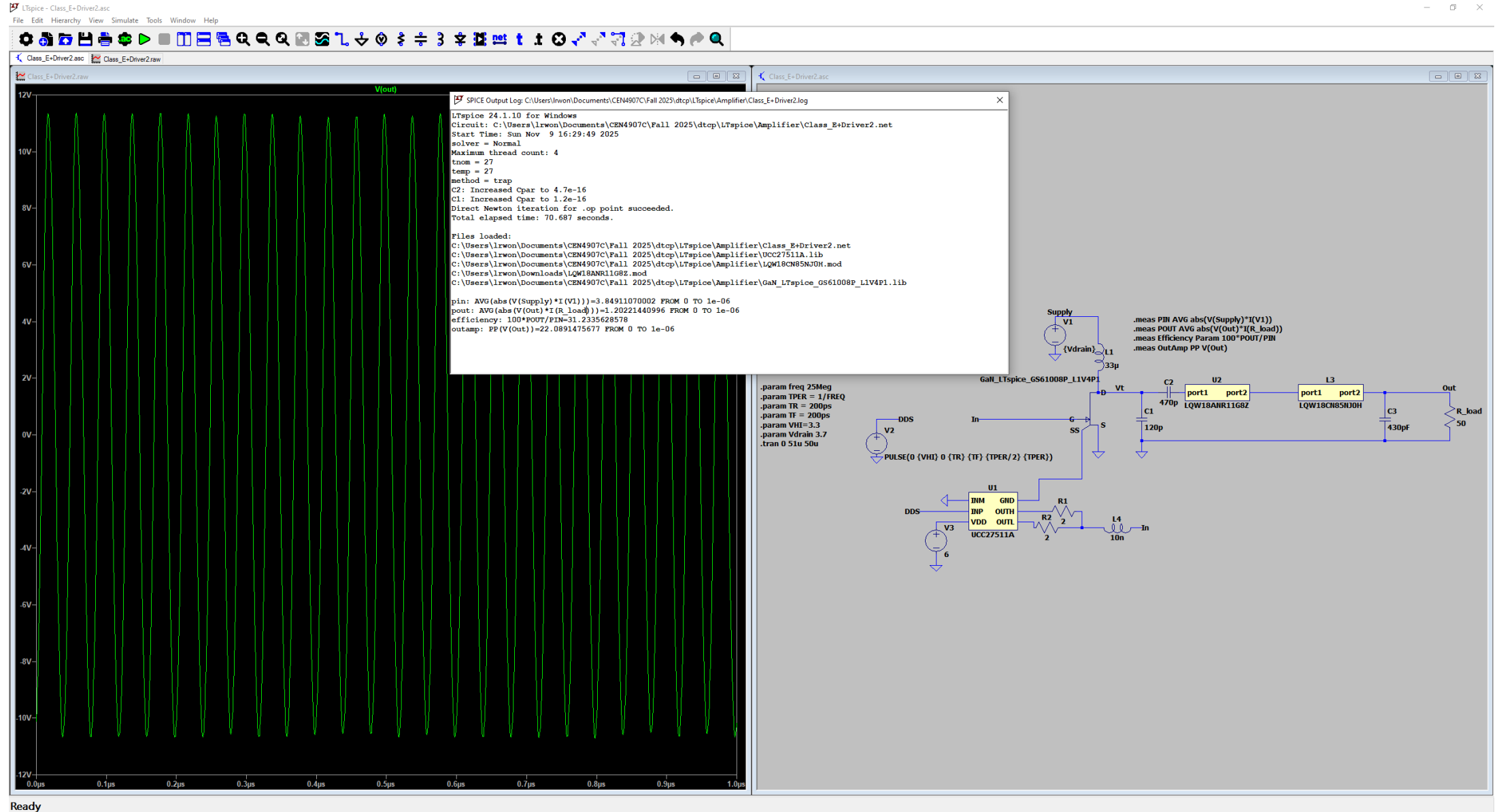
Source: <https://ludens.cl/Electron/RFamps/RFamps.html>

RF Power Amplification (Prior Art)

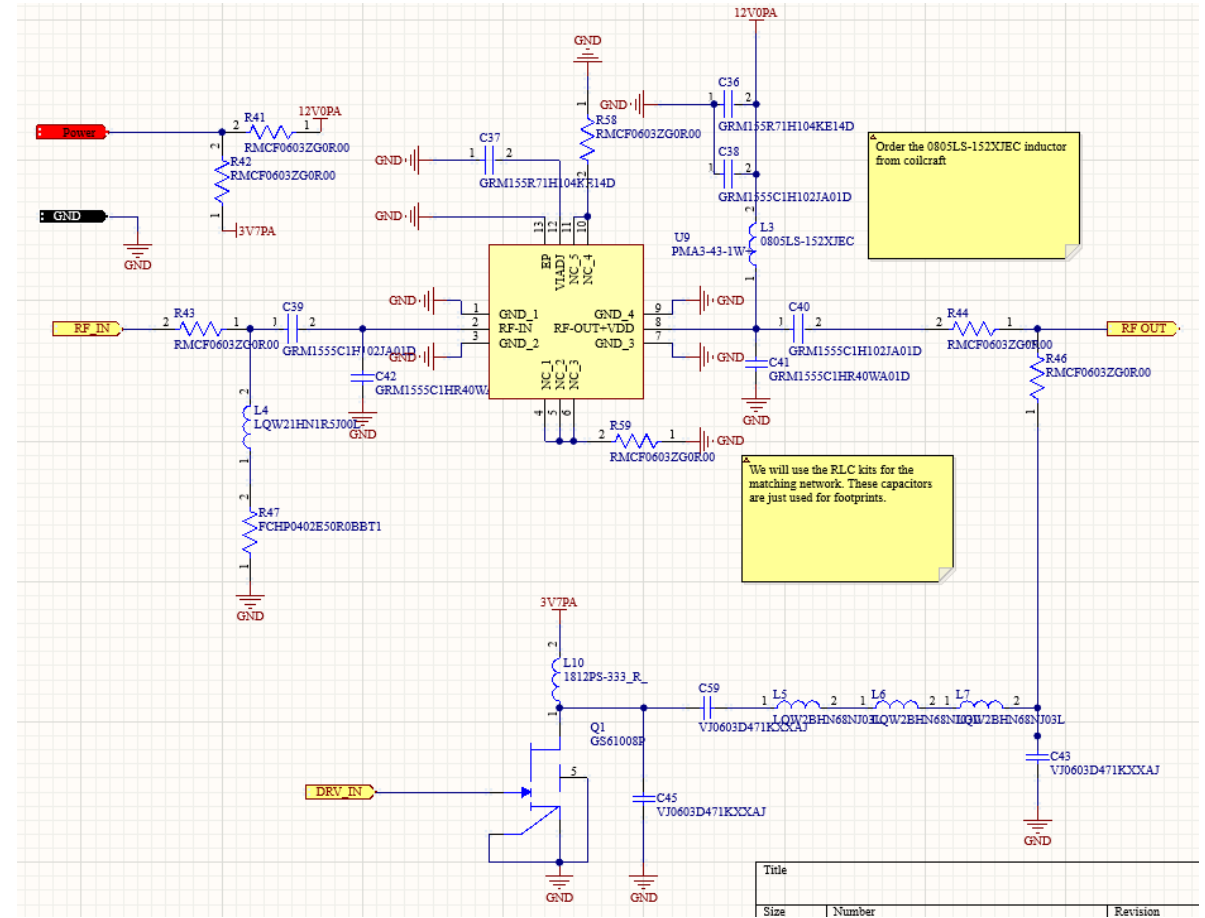
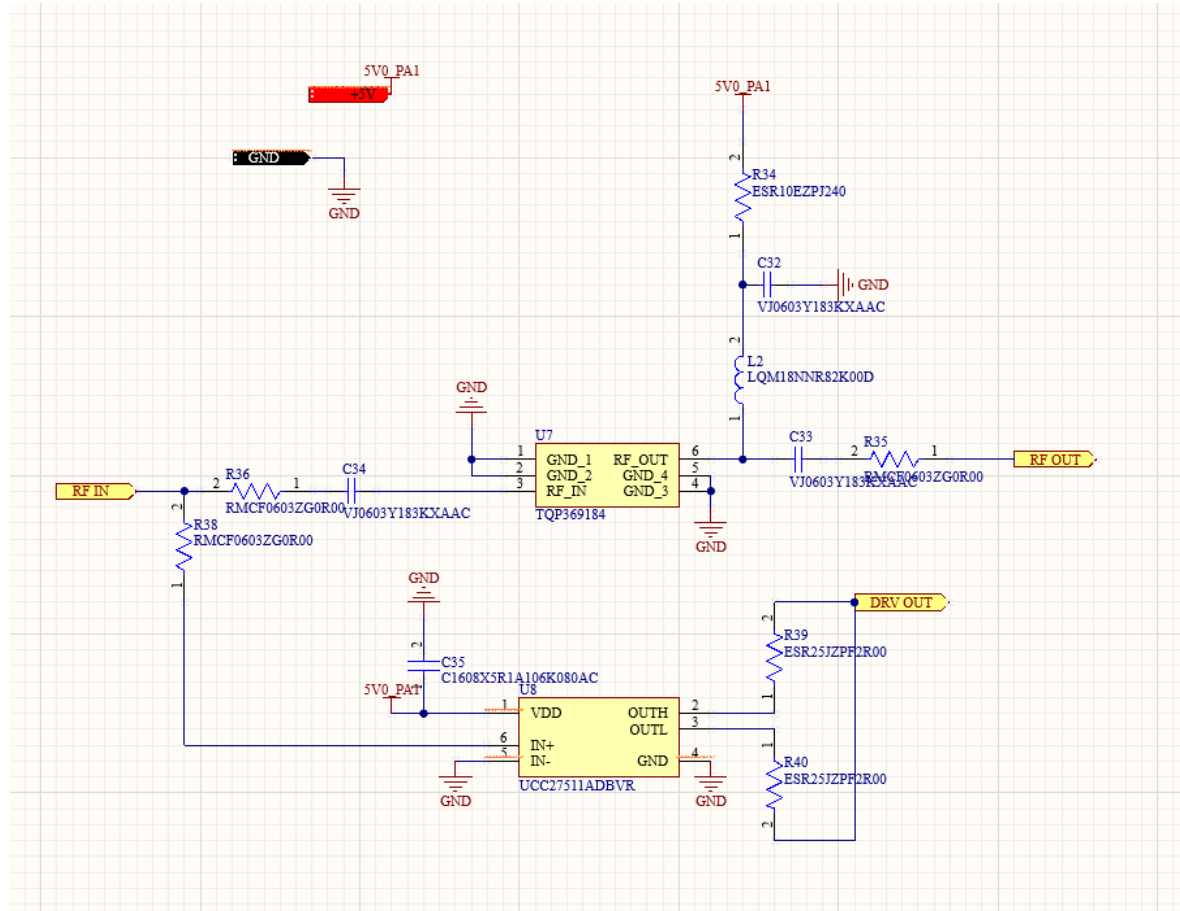


Source: H. Dadashzadeh, "Design and implementation of class-E power amplifiers and rectifiers for a dual-band capacitive wireless power transfer system," *University of British Columbia Library*, Oct. 2025, doi: <https://doi.org/10.14288/1.0406624>.

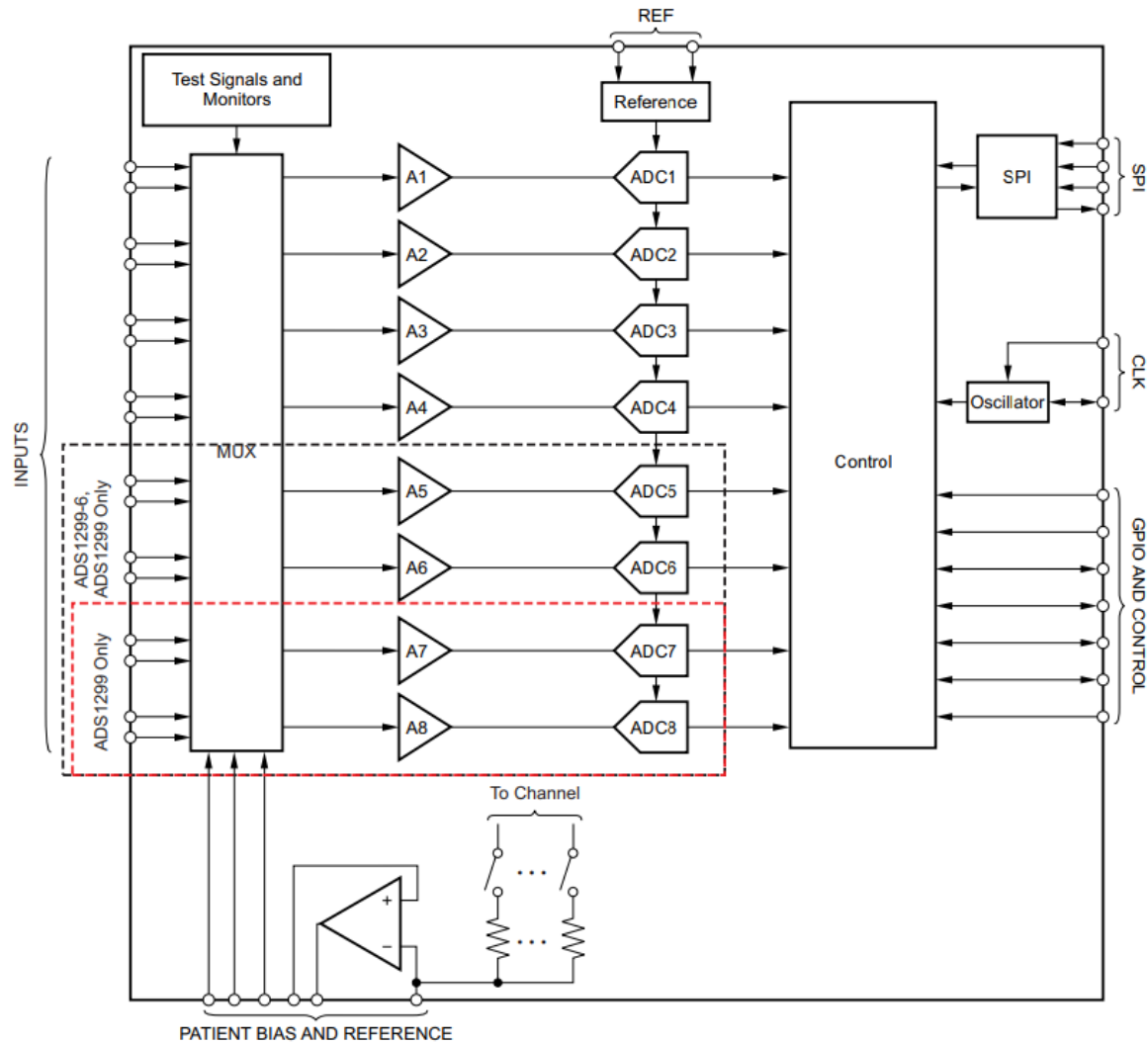
RF Power Amplification (Simulation)



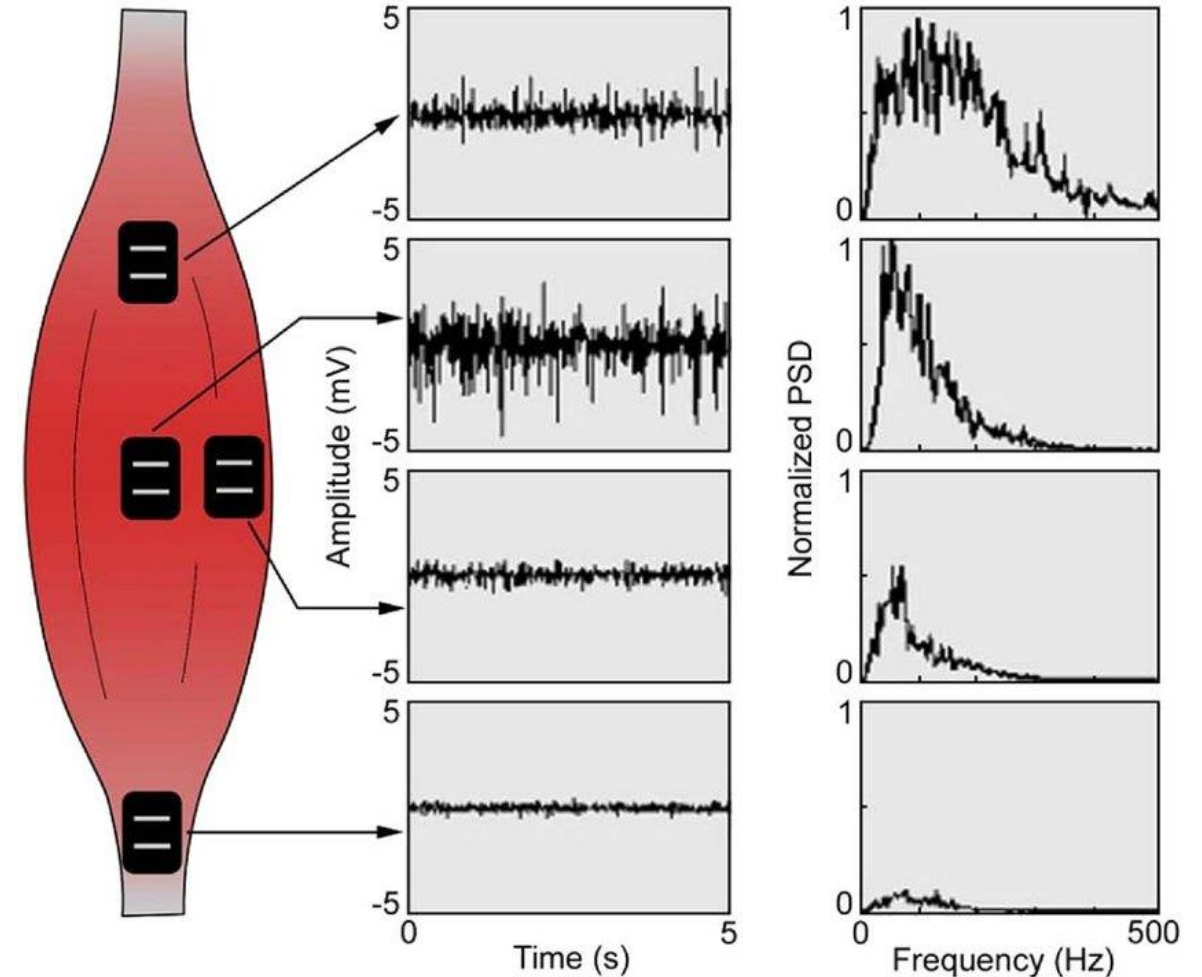
RF Power Amplification



EMG Analog Front-End (AFE)

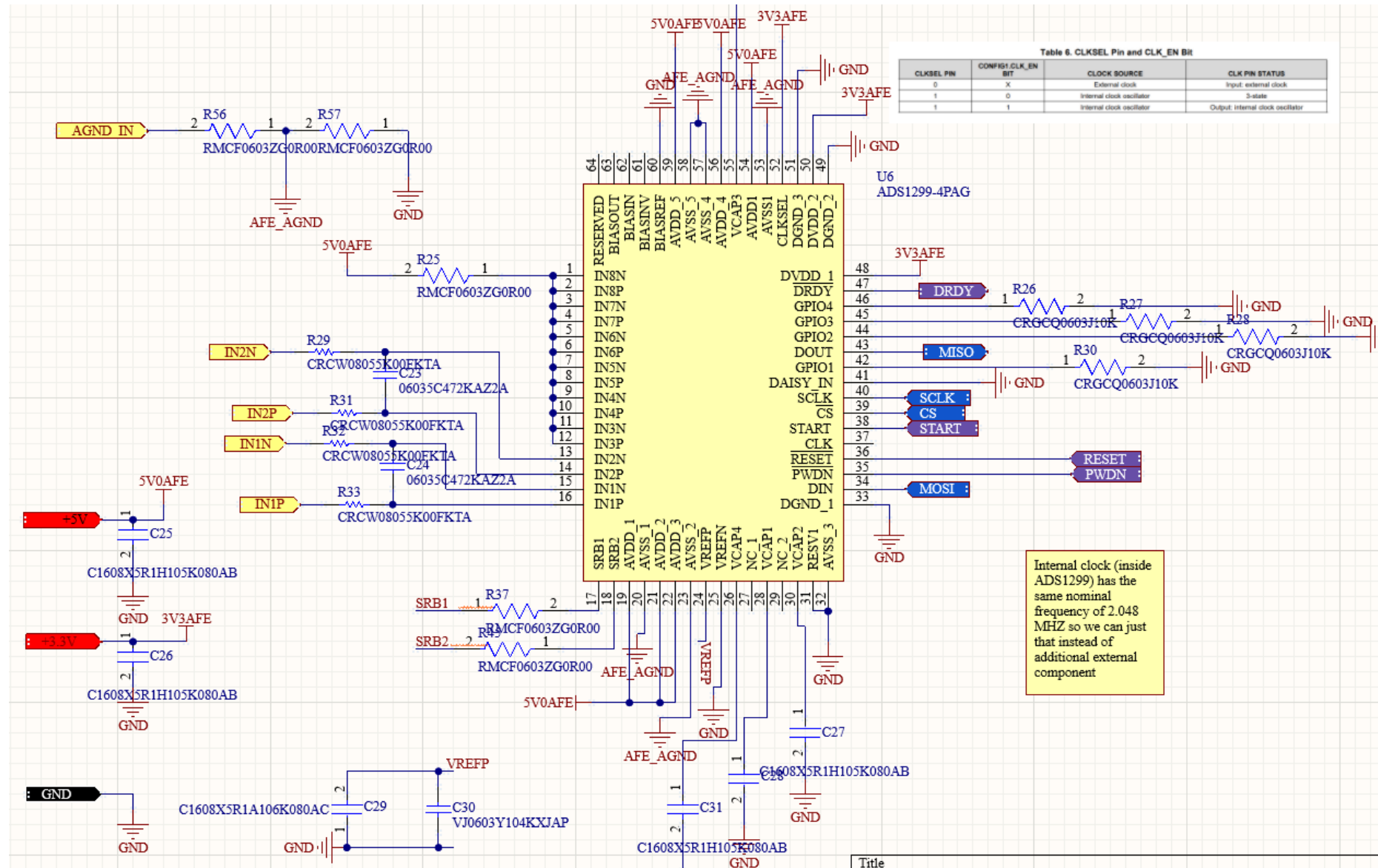


Source: Texas Instruments
<https://www.ti.com/product/ADS1299>

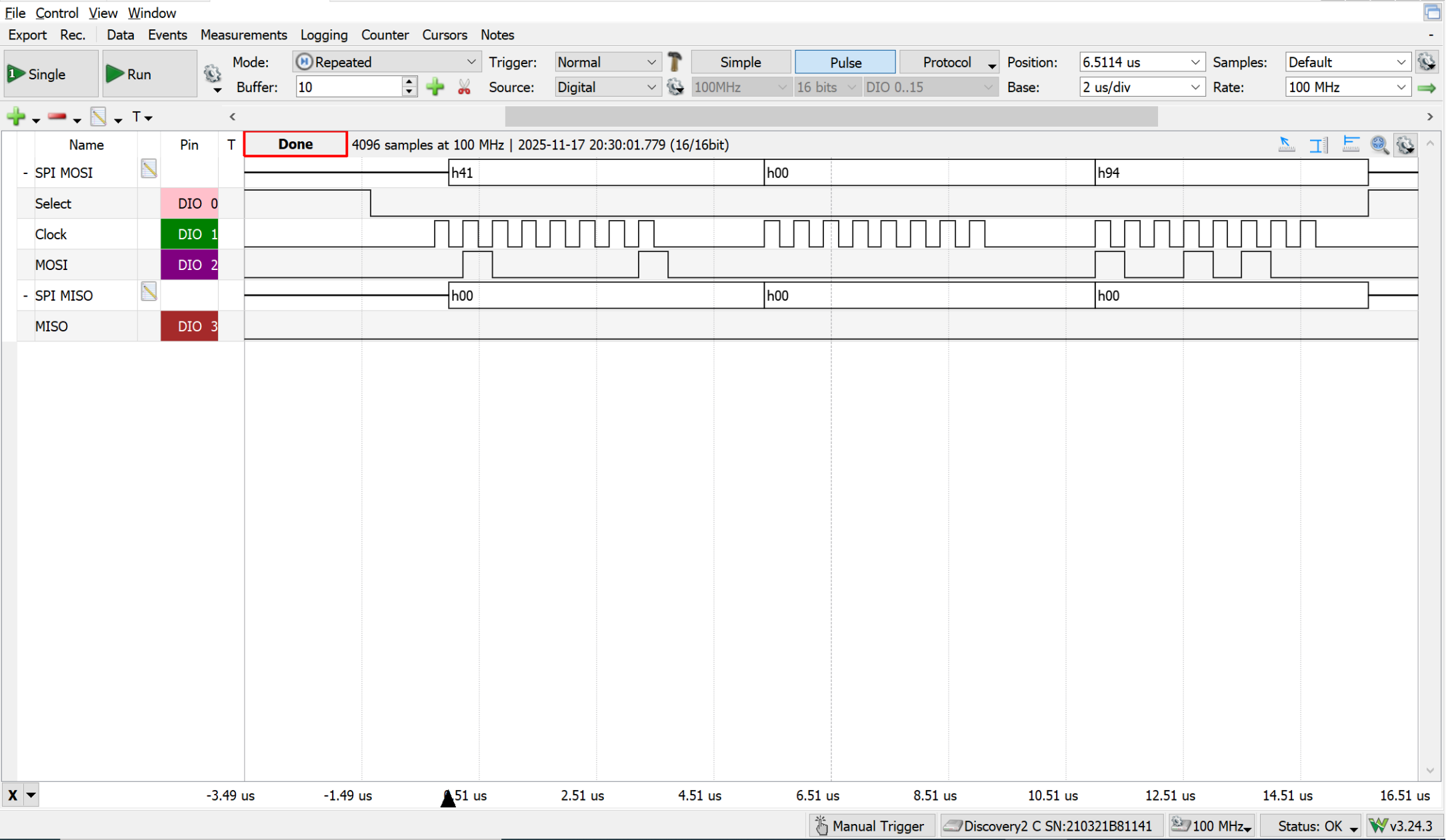


Source: Micera, Silvestro et al. (2010). Control of hand prostheses using peripheral information. IEEE Rev Biomed Eng 3:48-68. IEEE reviews in biomedical engineering. 3. 48-68. 10.1109/RBME.2010.2085429.

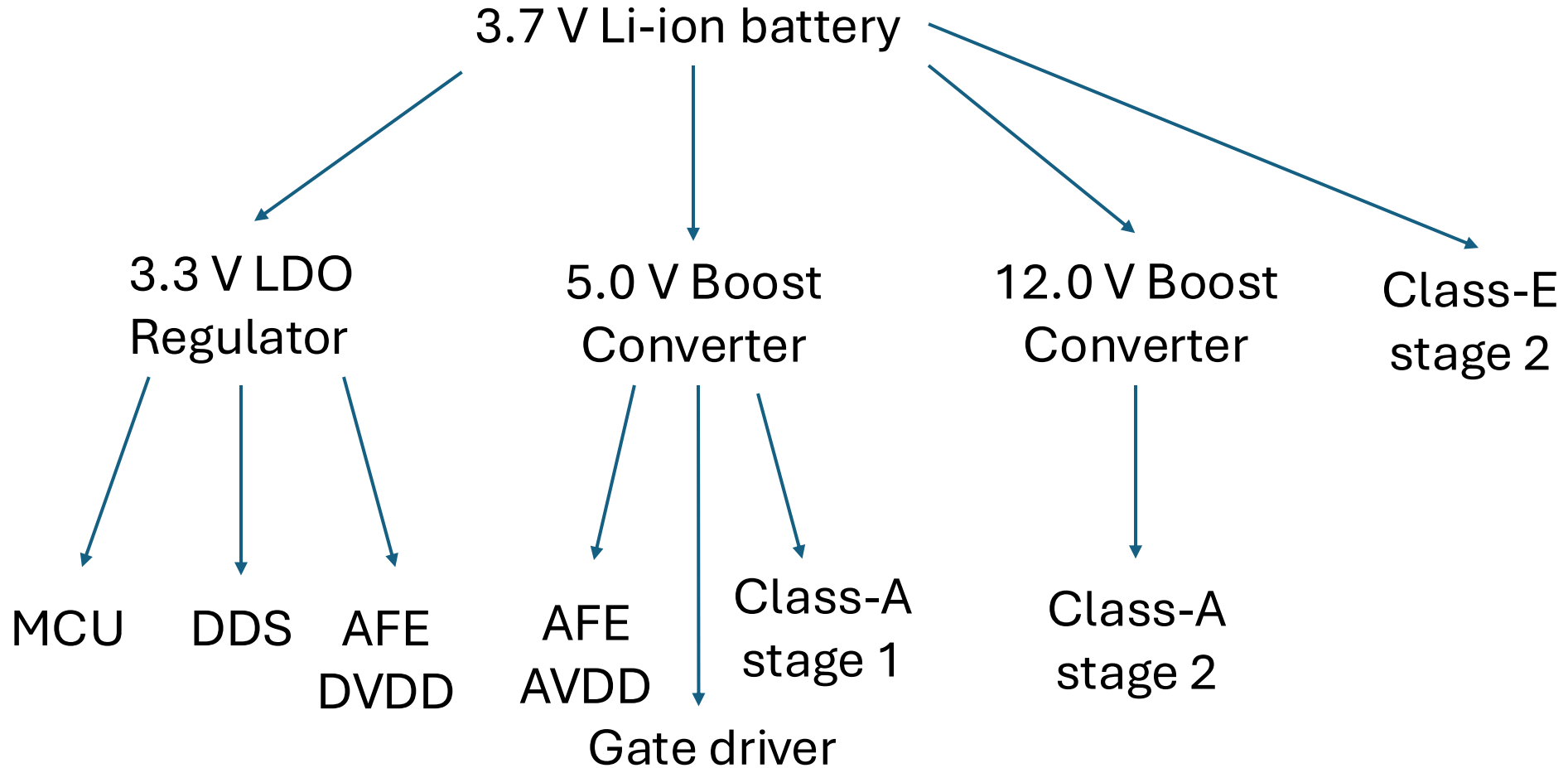
EMG Analog Front-End (AFE)



Firmware (SPI)



DC Power



Deliverable Plan

First Prototype

The first prototype aims to realize the most basic functionality: a 25 MHz differential output at 1 W.

Task	Finish By	Leader	Supporters	Comments	Status
Design the PA circuit	End of October 2025	Luis Wong	Jeremiah Dados		✓
Design the EMG circuit	End of October 2025	Dmytro Stavskyi	Jeremiah Dados		✓
Design the first prototype PCB in Altium	Mid-November 2025	Jeremiah Dados	Luis Wong, Dmytro Stavskyi		✓
Place the fabrication order	Mid-November 2025	Jeremiah Dados	Han Wu		✓
Receive the board and assemble it	End of November 2025	Jeremiah Dados	Luis Wong, Dmytro Stavskyi		🟡
Validate board outputs on lab equipment	Mid-December 2025	Jeremiah Dados	Luis Wong, Dmytro Stavskyi	Present functional prototype at end of semester	✗
Validate board functionality in animal experiments	End of January 2026	Han Wu	Jeremiah Dados, Luis Wong, Dmytro Stavskyi	Work resumes 2nd week of January after winter break	✗

- Verify outputs / debug using RF oscilloscope, Spectrum Analyzer
- Test EMG using saline and tissue phantoms
- Then move to animal experiments (rat surgery)

Deliverable Plan

Second Prototype

The second prototype addresses issues found in the first design and focuses on improving compactness, flexibility, and performance.

Task	Finish By	Leader	Supporters	Comments	Status
Revisions to the circuit design	Mid-February 2026	Luis Wong, Dmytro Stavskyi	Jeremiah Dados		✗
Design and order the PCB	End of February 2026	Jeremiah Dados	Luis Wong, Dmytro Stavskyi		✗
Place the fabrication order	End of February 2026	Jeremiah Dados	Han Wu		✗
Receive the board and assemble it	Mid-March 2026	Luis Wong, Dmytro Stavskyi	Jeremiah Dados		✗
Validate board outputs on lab equipment	End of March 2026	Jeremiah Dados	Luis Wong, Dmytro Stavskyi		✗
Validate board functionality in animal experiments	End of April 2026	Han Wu	Jeremiah Dados, Luis Wong, Dmytro Stavskyi		✗

- Verify outputs / debug using RF oscilloscope, Spectrum Analyzer
- Test EMG using saline and tissue phantoms
- Then move to animal experiments (rat surgery)

Prototype PCB

